

Homework — 1.3.2 Databases

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 1.3.2 lesson. Show your working.

Part A — This week's topic (~14 marks)

A sports club uses the following two tables. Primary keys are shown in bold.

Member(**MemberID** , FirstName, Surname, MembershipType) Booking(**BookingID** , MemberID, CourtNumber, BookingDate)

1. Identify the **foreign key** in this design and state which table and attribute it links to. **[2]** (AO2)

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2. Write an SQL query that returns the **FirstName** and **Surname** of every member whose **MembershipType** is 'Junior', sorted by **Surname** in **descending** order. **[3]** (AO2)

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3. Write an SQL statement that uses a **JOIN** to list each member's **Surname** together with the **CourtNumber** and **BookingDate** of every booking they have made. **[3]** (AO2)

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4. A mail-order company records each order as a single record, where one order can contain several products:

Order(OrderID, CustomerName, CustomerAddress, ProductID, ProductName, Quantity)

(a) Explain why this table is **not** in First Normal Form (1NF), and state what 1NF requires. [2] (AO1/AO2)

(b) After splitting into order lines keyed on (OrderID, ProductID), the attribute ProductName depends on ProductID alone — only part of the key. Name this type of dependency and state which normal form removes it. [2] (AO1)

5. The club's booking system processes payments as **transactions**. Explain the **Durability** property of ACID and why it matters if the server loses power. [2] (AO1/AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.3.1) A digital photograph is to be uploaded to a website. State whether **lossy** or **lossless** compression is the more suitable choice and give **one** reason. [2] (AO1/AO2)

7. (1.2.3) State what is meant by the **Waterfall** model of software development and give **one** drawback of it. [2] (AO1)

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8. (1.1.3) Explain **one** reason why an SSD might be chosen over a magnetic hard disk drive in a laptop.
[2] (AO1)

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